

APPRENTICE RECORD BOOK

WELDER



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Using the Record Book



Scan the QR code to visit
tradesecrets.alberta.ca/record-book.

- Watch a video tutorial.
- Read step-by-step instructions.



Affix Identification Label

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Important message



As part of the Apprenticeship Education Agreement, both you and your sponsor will ensure that activities listed in the record book are successfully demonstrated and documented.

Constructive and effective communication, coaching, and mentorship are important.

Individual roles and obligations are summarized throughout the record book. The following references are also available:

- The *Skilled Trades and Apprenticeship Education Act*;
- The *Apprenticeship Education and Industry Training Programs Regulation*;
- The Welder Curriculum Guide.



Definitions and obligations

Apprentice

An apprentice is an individual who is registered in an apprenticeship education program. The apprentice's responsibilities with this record book are to:

- Demonstrate core competences by performing work activities to a set standard, as evaluated by a mentor.
- Complete the required on-the-job training and work experience hours in their program.
- Submit the **End of Period Endorsements and Acknowledgements** and **Verification of On-the-job Instruction Hours** sections of this record book at the end of each period of their program.

Sponsor

A sponsor is a person or entity who, pursuant to an apprenticeship education agreement, provides an apprentice with on-the-job instructions required as part of an apprenticeship education program. The sponsor's responsibilities with this record book are to:

- Verify and endorse that a qualified mentor has assessed and endorsed the apprentice's competence for each period of their program.
- Verify and endorse the apprentice's on-the-job instruction hours for each period.
- Ensure that the apprentice submits the required portions of this record book at the end of each period.

Mentor

A mentor is an employer, provider of in-classroom instruction, or an individual designated as the representative of an approved sponsor who has **journeyperson certification** or is deemed **qualified** to ensure apprentices receive quality guidance, instructions, and assessments throughout the apprenticeship education agreement. The mentor's responsibilities with this record book are to:

- Provide the apprentice with instructions, guidance, and opportunities to develop the required competences.
- Evaluate the apprentice's work and endorse any competences met.

Keep your sponsorship information up to date

If you change sponsors during your active apprenticeship, please complete and submit the appropriate documentation.

The appropriate documentation and forms (the Request for Approval of Apprenticeship Education Agreement and Notice of Release forms) can be accessed through this QR code or found by visiting the Alberta Apprenticeship and Industry Training (AIT) website (tradesecrets.alberta.ca).



Submit your documents through your MyTradesecrets account.

Entrance requirements



Make sure you have met the entrance requirements to attend the classroom instruction portion of your Apprenticeship Education Program. You are required to attend classroom instruction every year, so make sure you are eligible when the time comes to register.

The minimum requirements to attend the classroom instruction portion of this apprenticeship program are:

- English 10-2, Math 10-3.
- OR a pass mark in all five Canadian General Educational Development (GED) tests or Canadian Adult Education Credential (CAEC) tests.
- OR an Apprenticeship and Industry Training entrance examination.

To increase your chances of success in classroom instruction for the Welder Apprenticeship Education Program, successful completion of English 30-2, Math 30-3, Physics 20 OR Chemistry 20 OR Science 20 are recommended.

The following categories are reflected in the entrance requirements:

Comprehension

Welders are required to comprehend and use written and/or electronic communications throughout their day to facility and complete required tasks and activities.

These communications may include work orders, manuals, customer interactions, emails, and procedural documents.

Numeracy

Welders are often required to interpret drawings, sketches and/or other technical documents, which may contain numerical data.

They may be required to use measurement tools and make precise calculations in their trade. They may also be required to understand how formulas, fractions, decimals, ratios, proportions, and/or percentages can be used and converted and/or manipulated from one form to another.

Science

Welders are often required to apply the scientific principles associated to rigging, hoisting, lifting, and/or climbing. They may also be required to apply their knowledge of the science associated with the expansion and contraction of materials.

Apprentice service checklist



Prior to registering for a period of classroom instruction, research the items on the following list on the Tradesecrets website, and take the necessary action.

Classroom Instruction

tradesecrets.alberta.ca/apprentice-services/attend-classroom-instruction/



Examinations

tradesecrets.alberta.ca/apprentice-services/prepare-for-an-exam/



Financial Support

tradesecrets.alberta.ca/apprentice-services/get-financial-support/



Records Maintenance

tradesecrets.alberta.ca/apprentice-services/maintain-your-records/



Progress tracker

Use the progress tracker to ensure that you have completed all requirements to progress in each period. This is **not** an official record of completion.

On-the-job (OTJ) instruction (hours) and competences can be completed separately.



Mark each box when the requirement has been completed.



Note: On-the-job (OTJ) hours can change.

Progress tracker	Period 1	Period 2	Period 3
OTJ Instruction required (total hours) 	1560 ☐	1560 ☐	1560 ☐
Minimum Competence(s) Required	9 of 12 ☐	9 of 12 ☐	9 of 12 ☐
Classroom Instruction (8 weeks)	Yes ☐	Yes ☐	Yes ☐

Welder competences

Your goal as an apprentice is to acquire the core competences of your trade, which are listed below.

Core Competences Section Titles

Foundational Skills, Job Responsibilities, and Procedures
Properties of Metals
Material Preparation, Handling and Identification
Drafting, Drawings and Specifications
Shielded Metal Arc Welding (SMAW)
Gas Tungsten Arc Welding (GTAW)
Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), and Metal Cored Arc Welding (MCAW) and Submerged Arc Welding (SAW)

You must successfully complete specific activities in these core areas to demonstrate your competency.

- **Competence(s) required for each period of your program are outlined on the following pages, along with their related core competence.**
- **Your mentor will assess your progress and endorse each competence you successfully complete throughout your program.**
- **Your sponsor will endorse your completion once your mentor has endorsed all required competence(s) for the period.**
- **An additional competence related to Foundational Skills, Job Responsibilities, and Procedures is required for all periods, as outlined below.**

All periods

Throughout all periods of your program, your mentor will assess your work to ensure you are meeting the competence below:

CORE COMPETENCE SECTION TITLE:

FOUNDATIONAL SKILLS, RESPONSIBILITIES, AND PROCEDURES

Competences	Mandatory / Optional	Mentor Endorsement
Use Personal Protective Equipment to an industry standard.	Mandatory	Meet this competence throughout all apprenticeship periods. It <u>DOES NOT</u> require a mentor's endorsement.

Period 1 competences

During your first period of the Welder apprenticeship education program, your mentor will assess your work for these core competences section titles:

- Foundational Skills, Safety and Procedures
- Properties of Metals
- Material Preparation, Handling and Identification
- Drafting, Drawings and Specifications
- Shielded Metal Arc Welding (SMAW)
- Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), and Metal Cored Arc Welding (MCAW) and Submerged Arc Welding (SAW)



Have your mentor provide their endorsement when you can perform the competences listed below to the standard indicated.

- **Complete a minimum of nine (9) competences, including all mandatory competences.**
- **Completing additional competences will strengthen your ability, but it is not required to meet the period requirement.**
- **You are not required to submit the mentor endorsed competences listed in this section to AIT.**

CORE COMPETENCE SECTION TITLE: FOUNDATIONAL SKILLS, SAFETY AND PROCEDURES

Competences	Mandatory / Optional	Mentor Endorsement
Use required PPE for tasks according to industry standards.	Mandatory	Name: _____ Date: _____
Use power tools according to industry standards.	Mandatory	Name: _____ Date: _____
Set-up, operate, and maintain oxyfuel equipment.	Optional	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
PROPERTIES OF METALS**

Competences	Mandatory / Optional	Mentor Endorsement
Explain heat input, polarity, arc characteristics and their impact on welds.	Optional	Name: _____ Date: _____
Identify and select filler metals accordingly.	Mandatory	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
MATERIAL PREPARATION, HANDLING AND IDENTIFICATION**

Competences	Mandatory / Optional	Mentor Endorsement
Prepare material surfaces for welding.	Mandatory	Name: _____ Date: _____
Use measuring tools to verify material dimensions.	Mandatory	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
DRAFTING, DRAWINGS AND SPECIFICATIONS**

Competences	Mandatory / Optional	Mentor Endorsement
Interpret welding symbols and joint types.	Optional	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
SHIELDED METAL ARC WELDING (SMAW)**

Competences	Mandatory / Optional	Mentor Endorsement
Perform SMAW welds.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:

**GAS METAL ARC WELDING (GMAW), FLUX-CORED ARC WELDING (FCAW),
AND METAL CORED ARC WELDING (MCAW) AND SUBMERGED ARC
WELDING (SAW)**

Competences	Mandatory / Optional	Mentor Endorsement
Perform GMAW welds.	Optional	Name: _____ Date: _____
Perform FCAW welds.	Optional	Name: _____ Date: _____
Perform MCAW welds.	Optional	Name: _____ Date: _____



Once you have completed a minimum of 9 of 12 competences, including all mandatory competences, complete the End of Period 1 endorsement and acknowledgement below with your current sponsor.

Submit this section to AIT and earn credit toward your apprenticeship education program.



SUBMIT to AIT

End of Period 1 Sponsor's (Business Name) Competence Endorsement:

I am satisfied that (a) qualified mentor(s) has assessed competence for the apprentice and that the mentor(s) has determined that the apprentice has demonstrated competency.

End of Period 1 Apprentice's Competence Acknowledgement:

I acknowledge that a minimum of 9 (nine) competences have been completed.

**Sponsor (Business Name)
Printed Name and Phone
Number**

**Sponsor
Endorsement**

**Apprentice
Acknowledgement**

Name:

Signature:

AIT Identifier:

Phone Number:

Date:

Signature:

Period 2 competences

During your second period of the Welder apprenticeship education program, your mentor will assess your work for these core competences section titles:

- Foundational Skills, Safety and Procedures
- Properties of Metals
- Material Preparation, Handling and Identification
- Drafting, Drawings and Specifications
- Shielded Metal Arc Welding (SMAW)
- Gas Tungsten Arc Welding (GTAW)
- Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), and Metal Cored Arc Welding (MCAW) and Submerged Arc Welding (SAW)



Have your mentor provide their endorsement when you can perform the competences listed below to the standard indicated.

- **Complete a minimum of nine (9) competences, including all mandatory competences.**
- **Completing additional competences will strengthen your ability, but it is not required to meet the period requirement.**
- **You are not required to submit the mentor endorsed competences listed in this section to AIT.**

CORE COMPETENCE SECTION TITLE:

FOUNDATIONAL SKILLS, SAFETY AND PROCEDURES

Competences	Mandatory / Optional	Mentor Endorsement
Troubleshoot equipment issues affecting weld quality and implement corrective actions.	Mandatory	Name: _____ Date: _____
Identify and mitigate risks from fumes or reactivity.	Mandatory	Name: _____ Date: _____
Demonstrates the ability to complete tasks as directed.	Mandatory	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
PROPERTIES OF METALS**

Competences	Mandatory / Optional	Mentor Endorsement
Identify ferrous and non-ferrous metals.	Mandatory	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
MATERIAL PREPARATION, HANDLING AND IDENTIFICATION**

Competences	Mandatory / Optional	Mentor Endorsement
Perform fit-up tasks to specification; verify dimensions and alignment.	Optional	Name: _____ Date: _____
Demonstrate the ability to cut materials.	Optional	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
DRAFTING, DRAWINGS AND SPECIFICATIONS**

Competences	Mandatory / Optional	Mentor Endorsement
Interpret drawings to identify weld types and joint configurations.	Optional	Name: _____ Date: _____

**CORE COMPETENCE SECTION TITLE:
SHIELDED METAL ARC WELDING (SMAW)**

Competences	Mandatory / Optional	Mentor Endorsement
Perform fillet and groove welds.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:
GAS TUNGSTEN ARC WELDING (GTAW)

Competences	Mandatory / Optional	Mentor Endorsement
Perform GTAW welds.	Optional	Name: _____ Date: _____
Troubleshoot and maintain GTAW equipment.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:
GAS METAL ARC WELDING (GMAW), FLUX-CORED ARC WELDING (FCAW), AND METAL CORED ARC WELDING (MCAW) AND SUBMERGED ARC WELDING (SAW)

Competences	Mandatory / Optional	Mentor Endorsement
Perform wire feed multi-position welds.	Optional	Name: _____ Date: _____
Troubleshoot the cause of weld faults.	Mandatory	Name: _____ Date: _____



Once you have completed a minimum of 9 of 12 competences, including all mandatory competences, complete the End of Period 2 endorsement and acknowledgement below with your current sponsor.

Submit this section to AIT and earn credit toward your apprenticeship education program.



SUBMIT to AIT

End of Period 2 Sponsor's (Business Name) Competence Endorsement:

I am satisfied that (a) qualified mentor(s) has assessed competence for the apprentice and that the mentor(s) has determined that the apprentice has demonstrated competency.

End of Period 2 Apprentice's Competence Acknowledgement:

I acknowledge that a minimum of 9 (nine) competences have been completed.

Sponsor (Business Name)
Printed Name and Phone
Number

Sponsor
Endorsement

Apprentice
Acknowledgement

Name:

Signature:

AIT Identifier:

Phone Number:

Date:

Signature:

Period 3 competences

During your third period of the Welder apprenticeship education program, your mentor will assess your work for these core competences section titles:

- Foundational Skills, Safety and Procedures
- Properties of Metals
- Material Preparation, Handling and Identification
- Drafting, Drawings and Specifications
- Shielded Metal Arc Welding (SMAW)
- Gas Tungsten Arc Welding (GTAW)
- Gas Metal Arc Welding (GMAW), Flux-Cored Arc Welding (FCAW), and Metal Cored Arc Welding (MCAW) and Submerged Arc Welding (SAW)



Have your mentor provide their endorsement when you can perform the competences listed below to the standard indicated.

- **Complete a minimum of nine (9) competences, including all mandatory competences.**
- **Completing additional competences will strengthen your ability, but it is not required to meet the period requirement.**
- **You are not required to submit the mentor endorsed competences listed in this section to AIT.**

CORE COMPETENCE SECTION TITLE:

FOUNDATIONAL SKILLS, SAFETY AND PROCEDURES

Competences	Mandatory / Optional	Mentor Endorsement
Participate in toolbox talks and mentor peers.	Mandatory	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:

PROPERTIES OF METALS

Competences	Mandatory / Optional	Mentor Endorsement
Apply preheat / interpass / post-weld heat treatment.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:

MATERIAL PREPARATION, HANDLING AND IDENTIFICATION

Competences	Mandatory / Optional	Mentor Endorsement
Execute complex layout/fit-up.	Optional	Name: _____ Date: _____
Apply quality control procedures and document inspection results.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:

DRAFTING, DRAWINGS AND SPECIFICATIONS

Competences	Mandatory / Optional	Mentor Endorsement
Adapt to drawing revisions and field conditions.	Mandatory	Name: _____ Date: _____
Interpret drawings and adhere to specifications.	Mandatory	Name: _____ Date: _____
Communicate drawing interpretation; document changes or discrepancies.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:

SHIELDED METAL ARC WELDING (SMAW)

Competences	Mandatory / Optional	Mentor Endorsement
Weld butt joints.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:
GAS TUNGSTEN ARC WELDING (GTAW)

Competences	Mandatory / Optional	Mentor Endorsement
Perform GTAW welds.	Optional	Name: _____ Date: _____

CORE COMPETENCE SECTION TITLE:
GAS METAL ARC WELDING (GMAW), FLUX-CORED ARC WELDING (FCAW), AND METAL CORED ARC WELDING (MCAW) AND SUBMERGED ARC WELDING (SAW)

Competences	Mandatory / Optional	Mentor Endorsement
Perform multi-process welds meeting code and NDE acceptance.	Optional	Name: _____ Date: _____
Set up equipment and verify parameters to maintain WPS compliance.	Optional	Name: _____ Date: _____
Complete a repair to meet final inspection.	Optional	Name: _____ Date: _____



Once you have completed a minimum of 9 of 12 competences, including all mandatory competences, complete the End of Period 3 endorsement and acknowledgement below with your current sponsor.

Submit this section to AIT and earn credit toward your apprenticeship education program.



SUBMIT to AIT

End of Period 3 Sponsor's (Business Name) Competence Endorsement:

I am satisfied that (a) qualified mentor(s) has assessed competence for the apprentice and that the mentor(s) has determined that the apprentice has demonstrated competency.

End of Period 3 Apprentice's Competence Acknowledgement:

I acknowledge that a minimum of 9 (nine) competences have been completed.

**Sponsor (Business name)
Printed Name and Phone
Number**

Name:

Phone Number:

**Sponsor
Endorsement**

Signature:

Date:

**Apprentice
Acknowledgement**

AIT Identifier:

Signature:

Verification of on-the-job instruction and work experience hours



- Record your on-the-job instruction hours and get your current sponsor’s contact information to validate them.
- Complete 1560 verified on-the-job hours for each period of your program, for a total of 4680.
- Submit your entries to earn credit toward your apprenticeship education program.
- If you are changing sponsors (business name), create an entry and ensure the required signature and contact information are included.

The initial time entry must be dated on or later than the Apprenticeship Education Agreement start date. Each subsequent entry must be after the last “End Date”.

Agreement Start Date: _____
 (NOTE: Use the Agreement Start Date found on the Apprentice Identification label)

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)			
Start Date:	End Date:	Hours Worked:	Sponsor's (Business Name) Information:
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
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			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
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			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
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On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

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			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

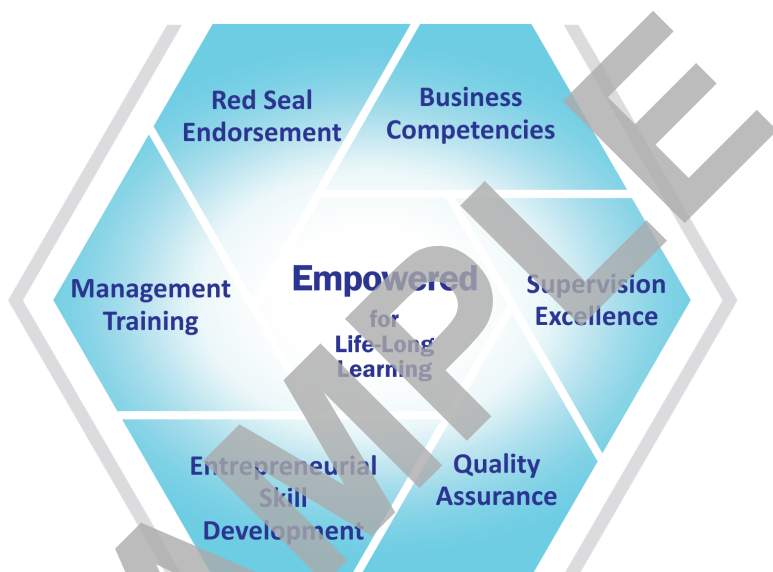
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

On-the-Job Instruction Hours (Minimum Hours Required: 4680 Hours)

Start Date: **End Date:** **Hours Worked:** **Sponsor's (Business Name) Information:**

			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____
			Sponsor (Business Name) Identification: _____ _____ Address: _____ Phone Number: _____

**Empowered
for life-long
learning**



For more information, visit the Alberta Careers, Learning, and Employment Information website @ alis.alberta.ca



Notes

SAMPLE

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